Quotation No : Q25000264

Date : 07-Jul-2025

Subros Limited
A20/1 SIPCOT,Indl GrowthCentre, Sriperumbudur Post, Kancheepuram
Chennai
TAMILNADU

Kind Attention : Mr./Ms. Vinoth Kumar

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

For Absstem Industrial Solutions Pvt. Ltd.

Pramod Kumar

Director

8447642325

www.absstem.com

Absstem Shield:

Premium Service Across India



Absstem Technologies, a leading gas technology provider and industrial and medical gas generator manufacturer has come up with a new initiative called **Absstem Shield**, under which Annual Maintenance Contracts (AMC) and Comprehensive Maintenance Contracts (CMC) will be offered for oxygen generators. Our Service Plans keep your nitrogen and oxygen generators in excellent shape.

Key Features

- Our AMC/CMC solutions work with **ANY** brand **PSA/VPSA Oxygen & Nitrogen** Generators
- We offer **on-site checkups** for your machines to access and understand your machine's requirement
- **CMC solutions** that look after your machines for as long as **10 years and beyond**
- **24 x 7** access to spare parts
- Quickest TAT (Turnaround time) in the industry.
- Constant touch of our team with your operator on **monthly** basis, ensuring highest uptime in the industry.
- Genuine spares with 6-month warranty
- We provide constant **monitoring** along with **monthly status updates**

Absstem Shield

Our in-house technical shield engineers are in constant contact with the operators every month making sure that the purity levels and output efficiencies are consistent with the requirement you've asked for. Our AMC plans work with ANY brand and either PSA or VPSA oxygen generators and PSA nitrogen generators. We make sure that every service during the AMC period is free-of-charge. If there is a need for part replacement, the only cost you would incur would be that of the spares. AMCs are open 24*7 and assure a timely service maintenance schedule. Our plans include AMCs that include constant monitoring, monthly status updates, on-site visits and one of the quickest turn-around times in the country.

Absstem Shield Premium

Our Comprehensive contracts include anything and everything that your gas generators would require to make sure the output and purity levels are consistent. The CMC plans work with ANY brand and include monthly routine check-ups, on-site deployment of our technical shield engineers, and replacement of consumables at no additional cost. As mentioned before, our CMC plans are open 24x7.

Corporate Address
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Work Address: Plot No. 222 & 223 Industrial Area
Puth Khurd, New Delhi, Delhi - 110039
Contact: 1800 3010 3394

Features	Shield	Shield Premium
Visit	✓	✓
Consumables	✗	✓
Flexible Payment Options	✓	✓
Genuine Parts	✓	✓
OTR (One-time Resolution)	✓	✓
Service History	✓	✓
24x7 Support	✓	✓
Training	✓	✓
Our Warranty on OEM equipment	✗	✓
Priority Status	✗	✓
Breakdown Visits	Limited	Unlimited
10-year warranty on Sieves	✓	✓

Terms And Conditions

- ### Our Responsibility

  - Check installation and ambient conditions.
  - Maintenance by service engineer based on checklist
  - Documentation of the maintenance results
  - Scheduling the maintenance dates.

- ### Client Responsibility

  - Repair and Maintenance carried out under this contract do not release the user from the prescribed operator level preventive control routines described in the service manual.
  - It is the user's responsibility to make the PSA plant available for preventive maintenance to Absstem Service Engineer on his scheduled visit. Otherwise, such a visit will be charged.
  - The suggestions/recommendations given by our service engineers from time to time on the operation of the PSA Gas Plant shall be implemented by the user.
  - It is the user's responsibility to arrange for inspection/maintenance/repair/ spare part of main incoming power, oxygen or nitrogen pipeline leakages or any other equipment beyond PSA Gas Plant maintained by supplier.

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- Conditions:

  - The Service contract does not apply to any damage caused due to mishandling, neglect, accident, improper application and any repair or maintenance done by an unauthorized person.
  - We are not liable for any damages arising from improper use of the equipment, or failure to follow adequate safety precautions.
  - Proper cleaning of the Air Compressor and PSA Gas Generator should be done as per the maintenance schedule provided to Client. If any damage or operation of equipment is affected due to not following the preventive maintenance schedule given to the Client.
  - Proper/continuous (without fluctuation) incoming power should be provided by the Client. Any equipment damaged due to power fluctuation is not covered under this contract.
  - Separate earthing for PSA Gas Plant and pipeline should be provided by the Hospital. If any equipment is damaged due to the non-availability of separate earthing would not be covered under CMC.
  - The Client should arrange for at least 48 hours of Gas backup after reporting the problem.
  - The service contract for PSA Gas Generators should not be broken under any condition. If the service contract is broken by the user, then the next contract would be made after inspection of the PSA Gas Generator (Chargeable). Any spares or maintenance required before the continuation of the maintenance contract should be done at extra costs if required and would be the sole decision of supplier. We are offering you service contract with an increment of 7% every year.
  - If any maintenance/repair of the PSA Gas Generator is done by an unauthorized person, Service Contract would be void with immediate effect. And another Contract needs to be done after inspection of the machine (chargeable) and it would be sole decision of supplier.
  - Any equipment not covered under contract but is a part of PSA Gas Generator, the supplier is not liable for any services however if required on sole discretion of supplier it can be done on chargeable basis.
  - 10 years warranty on sieves is only applicable if the equipment is covered under Absstem premium and Absstem Shield (recommended consumables are changed as per the schedule given by supplier).
  - Any damages incurred in major equipment due to shot circuit and malfunction are not covered such as motors, electricals and electronics etc.

A world-standard plan, with pan-India service.

Thank You

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nGEN POWERED BY ABSSTEM INDUSTRIAL

Industrial

PSA Nitrogen Generator

Unique Features of Absstem Nitrogen Generator nGEN SH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

Ambient Design Parameters

a.	Temperature	0 to 45 Deg. Celsius
b.	Relative Humidity	up to 80 % (Max.)
c.	Altitude	up to 1000 m above Sea level

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CIN: U29256HR2022PTC103387

Product Description

a. Product Name	High Purity PSA Nitrogen Gas Generator
b. PSA Module	**nGEN SH**
c. Purity Range	99% - 99.95%

General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA N_2 Generator
- Nitrogen Storage Tank



Air Compressor	An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air.
Refrigeration dryer	Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air.
Air Tank	Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system.
Multistage Filtration system	Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels.
PSA system	Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air.
Storage Tank	Industrial Grade **Nitrogen** produced from the PSA system is stored in this tank for continuous supply.
Oxygen Analyser	Nitrogen produced is monitored by an Oxygen analyser. In case, O_2 impurity goes above the specified level, it gives an alarm and switches the plant to vent mode.

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HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**



Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth

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Absstem Shield Premium

(Comprehensive Annual Maintenance Contract)



Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

1) Guaranteed assurance of plant performance.
2) Genuine spare parts and consumables being used to provide longevity and durability.
3) Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
4) 24 x 7 On-Call service support to our customers for any basic trouble shooting
5) Emergency visit to attend any unplanned breakdown to any location within 48 hours.
6) No additional surprise charges as all breakdown spares and consumables covered.
7) The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.

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Product Description

Product	Description	Make	Qty
Lubricated Booster	IR-15T2X20 Booster	ABSSTEM	1

Caption	Specification
Gas to be compressed	Nitrogen
Suction Pressure	3 to 4 Barg
Discharge pressure	8 to 70 Barg
Motor rating	15 KW
Voltage	415 V AC
Gas Flow	51 to 62 NM3/hr
Type	Air Cooled Lubricated Reciprocating Booster

Product	Description	Make	Qty
HP Storage Tank 65 Barg	1000 ltr HP Storage Tank 65 Barg High Pressure Storage Tank		1

Caption	Specification
Volume	1000 litres
Working Pressure	65 Barg
Hydrotest Pressure	107.25 Barg

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Product	Description	Make	Qty
Gas Storage Vessel	AI-1000L Storage Vessel		2

Caption	Specification
Water Volume	1000 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	6.5 M3
Corrosion Allowance	1.5 MM

Product	Description	Make	Qty
Gas Storage Vessel	AI-500L Storage Vessel		1

Caption	Specification
Water Volume	500 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	3.25 M3
Corrosion Allowance	1.5 MM

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Product	Description	Make	Qty
nGen Nitrogen Generator High Purity	nGEN010SH Nitrogen		1

Caption	Specification
Gas Generated	Nitrogen
Gas Pressure	upto 6.5 Barg
FLOW AT 99% PURITY	46 Nm3/Hr DRY AIR REQUIRED 60 CFM
FLOW AT 99.5% PURITY	39 Nm3/Hr DRY AIR REQUIRED 59 CFM
FLOW AT 99.9% PURITY	29 Nm3/Hr DRY AIR REQUIRED 55 CFM
FLOW AT 99.95% PURITY	26 Nm3/Hr DRY AIR REQUIRED 57 CFM
COMPRESSED AIR PRESSURE REQUIRED	7.5 Barg
RECOMMENDED NITROGEN STORAGE	1000 Liters
POWER SUPPLY	230 V/ 110 V 50/60 Hz Single Phase
RECOMMENDED AIR RECEIVER	500 Litres Part of Air System
INPUT AIR DEW POINT MONITOR	INCLUDED
CMS Warranty	10 Years*

Product	Description	Make	Qty
Compressed Air System	C015F0094 Air System		1

Caption	Specification
Air Compressor KW	15 KW
Air Compressor Make	Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand
Air Dryer	Included
Dry Air Storage	700 Liters Included
Interconnecting Piping	Included
Pre filter	Included
Dry Air Generated	94 CFM
Air Pressure	7.5 Barg
POWER SUPPLY	400 V 3 Phase 50 Hz

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Product	Description	Make	Qty
Gas Storage Vessel	AI-2000L Storage Vessel		1

Caption	Specification
Water Volume	2000 Litres
Gas Stored	Nitrogen/Oxygen
Design Standard	ASME Sec. VIII Div.1 Pressure Vessel Design
Working Pressure	8 Barg
Hydrotest Pressure	13.2 Barg
Accessories Included	Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments
Material of Construction	IS2062 Gr.B
Gas Stored Volume	13 M3
Corrosion Allowance	1.5 MM

Commercial Offer

S No	Item Code	Item Name	Description	HSN	Qty	Unit Price	Discount	GST(%)	Amount
1	ASH74	Vacuum Blower	Spares	84141000	1	3,06,250.00	56,250.00	18.00	2,50,000.00
2	ASH155	15TNLX125PSI 20HP	Spares	84149090	1	8,00,000.00	2,00,000.00	18.00	6,00,000.00
3	IR-15T2X20	IR-15T2X20	Lubricated Booster	8414	1	6,50,000.00	2,20,000.00	18.00	4,30,000.00
4	HPS15	1000 ltr HP Storage Tank 65 Barg	HP Storage Tank 65 Barg	73110090	1	4,50,000.00	1,25,000.00	18.00	3,25,000.00
5	ASH310	Refrigerent Dryer 1M3	Spares	84213990	1	1,20,000.00	40,000.00	18.00	80,000.00
6	AIS0108ST	AI-1000L	Gas Storage Vessel	731100	2	1,00,000.00	25,000.00	18.00	1,50,000.00
7	AIS0058ST	AI-500L	Gas Storage Vessel	731100	1	60,000.00	12,000.00	18.00	48,000.00
8	ASH311	Low Pressure and high Pressure Pipeline	Spares	84213990	1	2,00,000.00	1,10,000.00	18.00	90,000.00
9	nGEN010SH	nGEN010SH	nGen Nitrogen Generator High Purity	840510	1	11,48,000.00	1,05,000.00	18.00	10,43,000.00
10	ACSC015F	C015F0094	Compressed Air System	840510	1	8,20,000.00	2,14,000.00	18.00	6,06,000.00

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11	AIS0208ST	AI-2000L	Gas Storage Vessel	731100	1	1,60,000.00	50,000.00	18.00	1,10,000.00
12	ASH312	Heat Exchanger for 1m3 Capacity	Spares	84213990	1	40,000.00	0.00	18.00	40,000.00
13	ASH313	VFD 5 KW Auto	Spares	84213990	1	30,000.00	5,000.00	18.00	25,000.00
14	ASH314	Pressure Reducing Station	Spares	84213990	1	1,00,000.00	30,000.00	18.00	70,000.00
15	ASH220	FILTER FOR HIGH PRESSURE	Spares	84219900	1	1,00,000.00	15,000.00	18.00	85,000.00
16	ASH60	N2 Mass Flow Meter	Spares	902680	2	1,75,000.00	66,000.00	18.00	2,18,000.00

Sub Total	41,70,000.00
GST(%) @18.00	7,50,600.00
Grand Total ()	49,20,600.00

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Terms And Conditions

1.	EXCLUSIONS & CLIENTS SCOPE	All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required).
2.	SUPERVISION OF ERECTION & COMMISSIONING	There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you.
3.	P&F	3% Extra.
4.	Freight	Extra at Actuals.
5.	Insurance	In your Scope.
6.	Payment Terms	50% Advance with your Prestegious Purchase Order and Balance against PI.
7.	Delivery	6-8 Weeks from the PO and Advance. Rest can be confirmed for Availability.
8.	Warranty	We hereby confirm that, our equipment shall have Suppliers warranty for a period of 12 months from the date of Commissioning or 18 months from date of Dispatch whichever is earlier.
9.	GST	As per the Commercial Table.
10.	Remarks	.

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Pramod Kumar

Director

8447642325

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